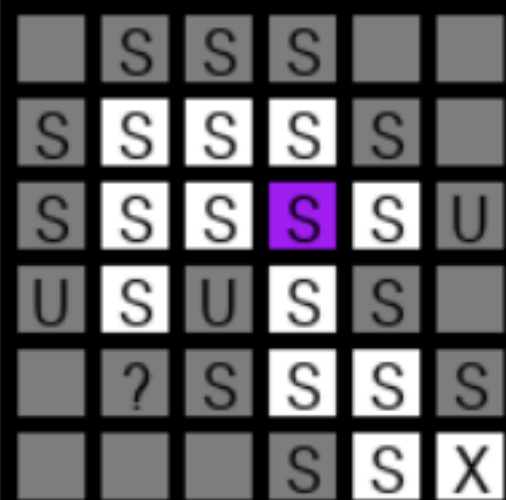




Player Coordinates: (5, 5)
Coins Grabbed: 2
Home is at (3, 2).
Agent detects a glitter. Player grabbed a coin.

Adventure World

CSINTSY Major

Course Output 2

ARCETA - CHAN - HONG - TAN

INTRODUCTION

Adventure World is a game inspired by Russell and Norvig's (2020) Wumpus World. The **Adventure World** is a $n \times n$ grid of cells, which some cells may include a Pit, from which the player cannot climb out. Some cells contain a gold coin. The player's goal is to collect at least two gold coins, then return home and depart. For this project, our goal is to **create an agent that tells the player which cells are safe, unsafe, and unknown**.

INFERENCE MODEL

The agent adds information about the player's adjacent locations to the knowledge base. The knowledge base will be put into the back end and be used to infer whether locations are safe, unknown, or unsafe. The cells are represented using cell(X, Y, S), in which X and Y represent the position of the cell and S is the status.

For the cell to be **safe**, it should either have no breeze nearby or if there is a breeze. The adjacent cells are without a breeze.

A fact to take note of is that the agent perceives a breeze whenever there is a pit adjacent to the player. So, for the cell to be marked **unsafe**, the knowledge-based must have sufficient knowledge that breeze is coming into that cell.

The cell will be considered **unknown** if the knowledge base does not have enough information to determine that the cell is a pit (e.g., the agent perceives a breeze, but which cell is the breeze coming from?)

CHALLENGES

Making sure that the knowledge base covers all the necessary cases was also a challenge.

There were occurrences during testing and debugging where all outputs were deemed the same (all unsafe, all safe, or all unsure, etc.)

Synchronization of the knowledge base information to the user interface display.

EVALUATION

The agent works on the maps that we have dynamically generated, properly asserting new information to the knowledge base and inferring pits. It was able to provide adequate, useful information on the maps generated by our program. It **can successfully handle 4 <= n maps** consistently, so **the bot performs well within expectations**. However, in a scenario where a cell has multiple pits near each other without any way to explore the other side, the status will remain unknown.

CONCLUSION

This project demonstrated the effective application of AI concepts to build an intelligent agent that aids puzzle solving. The inference engine's logical reasoning enabled safe navigation and enhanced the player's decision-making. By utilizing the knowledge base, the agent effectively avoided dangers and collected gold coins.